Chapter 21: Finer Monotone Properties

Convex Analysis and Monotone Operator Theory (2nd ed., 2017)

Bauschke & Combettes

Lecture Notes

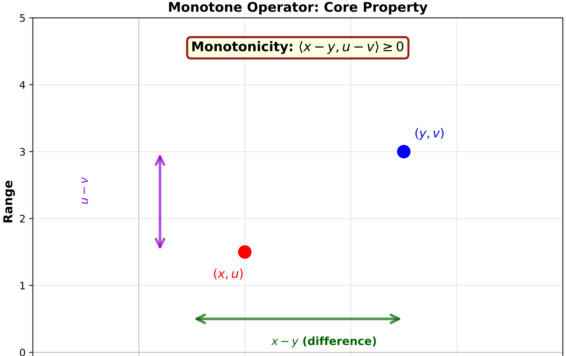
Outline I

Core Concepts

- **Monotone Operator**: $A : H \to 2^H$ is monotone if

$$\langle x - y, u - v \rangle \geq 0 \quad \text{for all } (x,u),(y,v) \in \operatorname{gra} A$$

- **Maximally Monotone**: A monotone operator A such that no proper monotone extension exists
- **Graph**: $\operatorname{gra} A = \{(x,u) \mid u \in A(x)\}$
- **Domain**: $\operatorname{dom} A = \{x \in H \mid A(x) \neq \emptyset\}$
- **Range**: $\operatorname{ran} A = \{u \in H \mid \exists x : u \in A(x)\}$



Theorem 21.1: Maximal Monotonicity Characterization

Theorem (Theorem 21.1)

Let $A : H \to 2^H$ be monotone. Then A is maximally monotone if and only if

$$0 \in ran(A + Id).$$

Proof Sketch.

- $\Rightarrow$ If A is maximally monotone, then $\mathrm{ran}(A + \mathrm{Id}) = H$
- $\Leftarrow$ If $0 \in \mathrm{ran}(A + \mathrm{Id})$ for all monotone extensions, maximality follows from the equivalence to Proposition 20.56

□

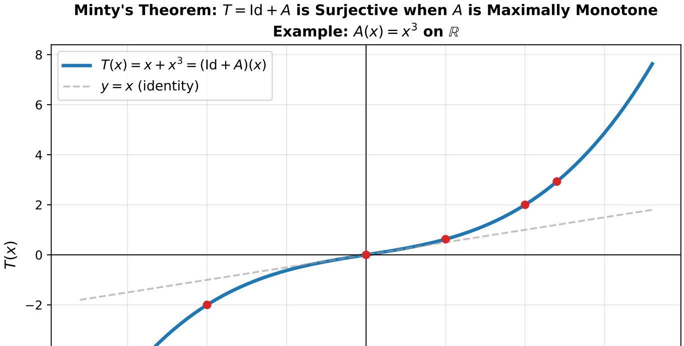
Consequence: For maximally monotone A: $\boxed{\mathrm{ran}(A + \mathrm{Id}) = H}$

Minty's Theorem: Intuition and Example

Definition (Minty's Theorem)

If $A : H \to 2^H$ is maximally monotone, then the operator $T = \mathrm{Id} + A$ is surjective: $\mathrm{ran}(T) = H$.

Example 21.3: Let $\mathcal{K}$ be a real Hilbert space, $L \in \mathcal{B}(H, \mathcal{K})$. Set $T = \mathrm{Id} + L^* L$ and $R = \mathrm{Id} + LL^*$.
Result: $T : H \to H$ and $R : \mathcal{K} \to \mathcal{K}$ are bijective with continuous inverses (from Theorem 21.1 and properties of $L^* L$).



Applications of Minty's Theorem

Theorem (Theorem 21.2)

Let $f \in \Gamma_0(H)$. Then ∂f is maximally monotone.

Proof: Combine Example 20.3, Proposition 16.45, and Theorem 21.1.

Proposition (Proposition 21.4)

Let $H = L^2([0, T]; H)$ and let A be the time-derivative operator:

$$A(x) = \begin{cases} \{x'\} & \text{if } x \in W^{1,2}([0, T]; H) \text{ and } x(0) = x_0 \\ \emptyset & \text{otherwise} \end{cases}$$

Then A is maximally monotone.

Proposition (Proposition 21.5)

Let C be a nonempty compact convex subset of H. Then $-\phi_C$ is maximally monotone, where ϕ_C is the farthest-point operator.

Theorem 21.8: The Debrunner-Flor Separation Theorem

Theorem (Debrunner-Flor Theorem)

Let $A : H \to 2^H$ be a monotone operator such that gra $A \neq \emptyset$. Then

$$(\forall w \in H) \quad \left(\exists x \in \overline{\text{conv dom } A}\right) \quad 0 \leq \inf_{(y,v)\in \text{gra } A} \langle y - x \mid v - (w - x)\rangle.$$

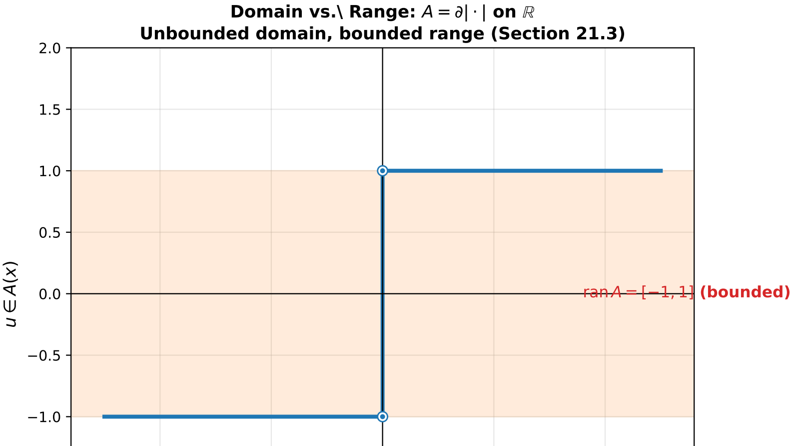
Key Consequence:

Theorem (Theorem 21.9: Maximal Monotone Extension)

Let $A : H \to 2^H$ be monotone. Then there exists a maximally monotone extension $\bar{A}$ of A such that dom $\bar{A} \subseteq \overline{\text{conv dom } A}$.

Debrunner-Flor: Proof and Intuition

- The Debrunner-Flor theorem provides a **separation property** for monotone sets
- For every point $w \in H$, there exists a point x in the closure of the convex hull of the domain such that the infimum of inner products is non-negative
- This is used to show existence of maximally monotone extensions



Definition 21.10: Local Boundedness

Definition (Local Boundedness)

Let $A : H \to 2^H$ and let $x \in H$. Then A is **locally bounded at** x if there exists $\delta \in \mathbb{R}_{++}$ such that $A(B(x;\delta))$ is bounded.

Proposition (Proposition 21.11)

Let $A : H \to 2^H$ be monotone, set $Q_1 : H \times H \to H : (x, u) \mapsto x$, and suppose that $z \in \operatorname{int} Q_1(\operatorname{dom} F_A)$. Then A is locally bounded at z.

Interpretation: Interior points of the domain projection are points of local boundedness

Proposition 21.12: Domain Relations I

Proposition (Proposition 21.12)

Let $A : H \to 2^H$ be maximally monotone and set $Q_1 : H \times H \to H : (x, u) \mapsto x$. Then

$$\mathrm{int}\ \mathrm{dom}\ A \subset \mathrm{int}\ Q_1(\mathrm{dom}\ F_A) \subset \mathrm{dom}\ A \subset Q_1(\mathrm{dom}\ F_A) \subset \overline{\mathrm{dom}\ A}.$$

Consequently, $\mathrm{int}\ \mathrm{dom}\ A = \mathrm{int}\ Q_1(\mathrm{dom}\ F_A)$ *and* $\mathrm{dom}\ A = Q_1(\mathrm{dom}\ F_A)$.
If $\mathrm{int}\ \mathrm{dom}\ A \neq \emptyset$, *then* $\mathrm{int}\ \mathrm{dom}\ A = \overline{\mathrm{dom}\ A}$.

Proposition 21.12: Domain Relations II

Example (Example 21.13)

Let C be a nonempty closed convex subset of H and set $Q_1 : H \times H \to H : (x, u) \mapsto x$. Then $Q_1(\text{dom } F_{N_C}) = C$.

Corollary (Corollary 21.14: Convexity of Domain and Range)

Let $A : H \to 2^H$ be maximally monotone, and set Q_1 and Q_2 as above. Then

$$\text{dom } A = Q_1(\text{dom } F_A), \quad \text{int dom } A = \text{int } Q_1(\text{dom } F_A),$$

$$\overline{\text{ran } A} = Q_2(\text{dom } F_A), \quad \text{int ran } A = \text{int } Q_2(\text{dom } F_A).$$

Consequently, the sets dom A, int dom A, $\overline{\text{ran } A}$, and int ran A are convex.

Bunt-Kritikos-Motzkin Result

Corollary (Corollary 21.16: Bunt-Kritikos-Motzkin)

Suppose that H is finite-dimensional and let C be a Chebyshev subset of H. Then C is closed and convex.

Proof: Uses Remark 3.11(i), Example 20.33, and Corollary 21.14 applied to the farthest-point operator.

Consequence: In finite dimensions, Chebyshev sets (closest-point unique) must be closed and convex.

Proposition (Proposition 21.17)

Let $A : H \to 2^H$ *be maximally monotone and let* $x \in \text{dom } A$*. Then* $\text{rec}(Ax) = N^x_{\overline{\text{dom } A}}$.

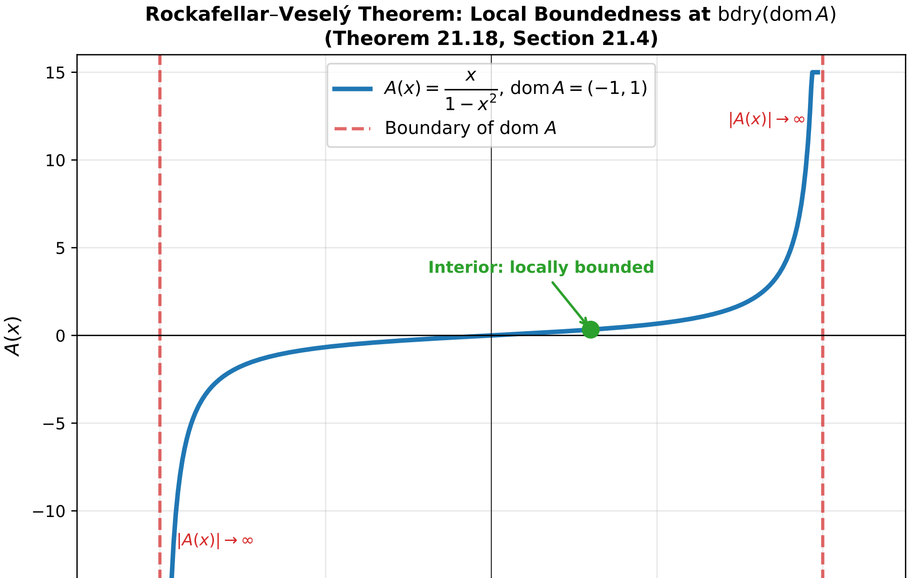
Interpretation: The recession cone of $A(x)$ equals the normal cone to the closure of the domain at x.

Theorem (Theorem 21.18: Rockafellar-Vesely)

Let $A : H \to 2^H$ *be maximally monotone and let* $x \in H$*. Then* A *is locally bounded at* x *if and only if* $x \notin \text{bdry dom } A$.

Key Fact: Local boundedness characterizes interior points of the domain.

Rockafellar-Vesely Example



Surjectivity Corollaries I

Corollary (Corollary 21.19)

Let $A : H \to 2^H$ be monotone, and suppose that C is a compact subset of int dom A. Then $A(C)$ is bounded.

Corollary (Corollary 21.20)

Let $A : H \to 2^H$ be maximally monotone. Then A is locally bounded everywhere on H if and only if dom $A = H$.

Surjectivity Corollaries II

Corollary (Corollary 21.21)

Let $A : H \to 2^H$ be maximally monotone and at most single-valued. Then A is strong-to-weak continuous everywhere on $\operatorname{int} \operatorname{dom} A$.

Key Results on Surjectivity:

Corollary (Corollary 21.23)

Let $A : H \to 2^H$ be maximally monotone. Then A is surjective if and only if A^{-1} is locally bounded everywhere on H.

Surjectivity Corollaries III

Corollary (Corollary 21.24)

Let $A : H \to 2^H$ be a maximally monotone operator such that

$$\lim_{\|x\| \to +\infty} \inf \|Ax\| = +\infty.$$

Then A is surjective.

Corollary (Corollary 21.25)

Let $A : H \to 2^H$ be maximally monotone with bounded domain. Then A is surjective.

Application to Convex Functions

Corollary (Corollary 21.26)

Let $f \in \Gamma_0(H)$ and let C be a compact subset of int dom f. Then $\partial f(C)$ is bounded.

Consequence: Subdifferentials of convex functions are locally bounded on the interior of their domains.

Theorem 21.27: Kenderov's Theorem I

Theorem (Kenderov's Theorem - Theorem 21.27)

Let $A : H \to 2^H$ be a maximally monotone operator such that int dom $A \neq \emptyset$. Then there exists a subset C of dom A that is a dense G_δ subset of int dom A such that, for every point $x \in C$:

1. *Ax is a singleton*
2. *Every selection of A is continuous at x*

Key Implications:

- On a dense subset, the multi-valued operator becomes single-valued
- Every selection (any choice of element from $A(x)$) is continuous at those points
- This is a generic property (holds on a dense G_δ set)

Theorem 21.27: Kenderov's Theorem II

Proof Strategy:

- Use a sequence $(\varepsilon_n)_{n\in\mathbb{N}}$ converging to 0
- Define sets C_n by requiring $y \in C_n$ iff diameter of $A(V) < \varepsilon_n$ for some open neighborhood V of y
- Show each C_n is open in int dom A
- Show $C = \bigcap_{n\in\mathbb{N}} C_n$ is the required dense G_δ set
- On C, the operator is single-valued and every selection is continuous

Application to Convex Functions: Fréchet Differentiability

Corollary (Corollary 21.28)

Let $f \in \Gamma_0(H)$ be such that int dom $f \neq \emptyset$. Then there exists a subset C of int dom f such that C is a dense G_δ subset of dom f, and such that f is Fréchet differentiable on C.

Interpretation:

- Convex functions with nonempty interior domain are generically Fréchet differentiable
- Differentiability holds on a dense G_δ subset (measure-theoretically "most" points)
- This is the sharpest result possible: there exist convex functions with no point of Gâteaux differentiability (see Example 21.6)

Generic Property: The set of points where Fréchet differentiability holds is not just nonempty—it is dense and of full "category" in the Baire sense.

Example 21.6: A Convex Function with Limited Differentiability

Example (Example 21.6)

Let $\mathbb{P} = \mathbb{N} \setminus \{0\}$, $H = \ell^2(\mathbb{P})$, let $(e_n)_{n\in\mathbb{P}}$ be the canonical orthonormal basis, and set

$$f : H \to [-\infty, +\infty] : x \mapsto \max\left\{1 + \langle x, e_1\rangle, \sup_{2\leq n\leq N} \langle x, \sqrt{n} e_n\rangle\right\}.$$

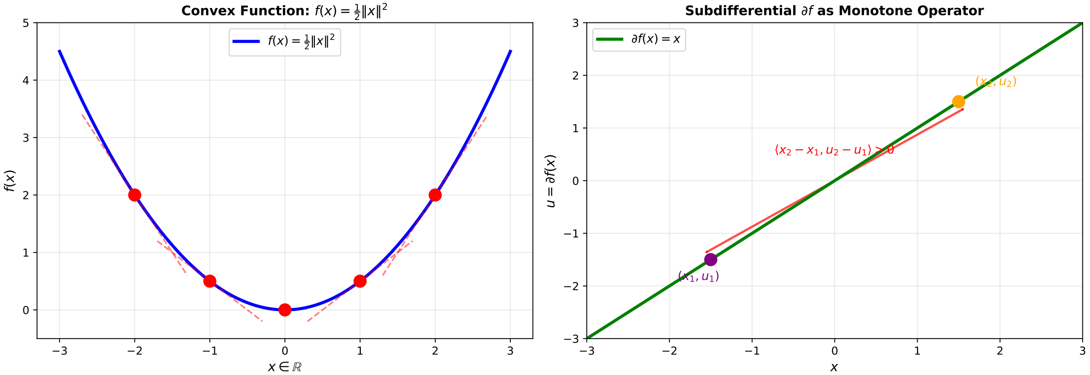
Then $f \in \Gamma_0(H)$ and ∂f is maximally monotone. However, gra ∂f is not closed in $H^{\text{strong}} \times H^{\text{weak}}$.

Note: This example shows that even though ∂f is maximally monotone, its graph may not be closed when considering different topologies.

Python Implementation: Monotone Operator Analysis

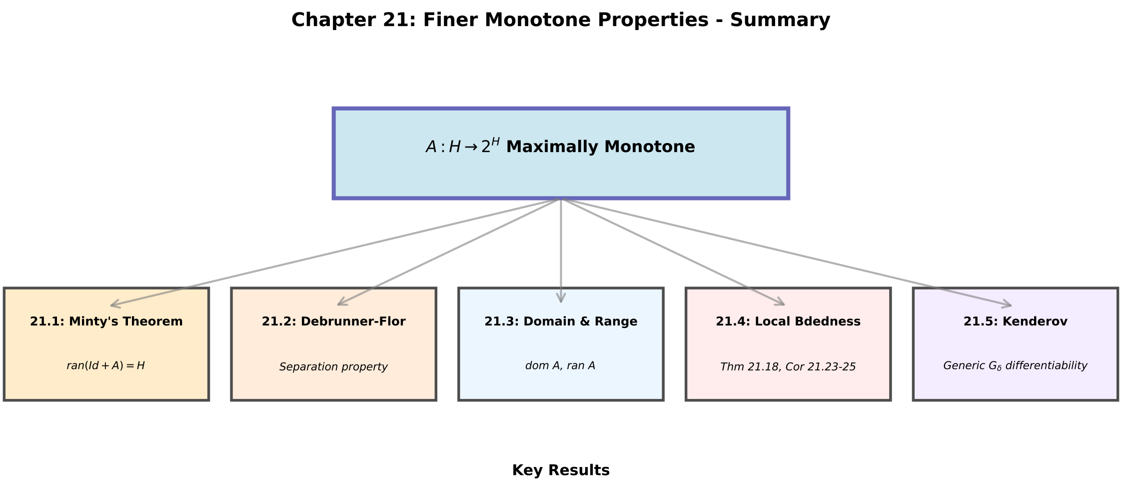
```python
import numpy as np
from scipy.optimize import fminbound

# Example: A(x) = subdifferential of f(x) = ||x||^2/2
class SubdifferentialNorm:
    def f(self, x):
        """Convex function: f(x) = 0.5 * ||x||^2"""
        return 0.5 * np.sum(x**2)

    def A(self, x):
        """Subdifferential: partial f(x) = x (single-valued)"""
        return x

    def verify_monotonicity(self, x1, x2):
        """Verify: <x1-x2, A(x1)-A(x2)> >= 0"""
        u1, u2 = self.A(x1), self.A(x2)
        return np.dot(x1 - x2, u1 - u2) >= -1e-10

# Test monotonicity
A = SubdifferentialNorm()
x1 = np.array([1.5, 2.0])
x2 = np.array([0.5, 1.0])
is_monotone = A.verify_monotonicity(x1, x2)
print(f"Monotone: {is_monotone}")  # True
print(f"Inner product: {np.dot(x1-x2, A(x1)-A(x2)):.4f}")
```

Visualization of Example Operator



The subdifferential ∂f of the convex function $f(x) = \frac{1}{2}\|x\|^2$ is the linear monotone operator $x \mapsto x$.

Summary: Five Main Themes



Key Theorems and Results I

1. **Theorem 21.1**: Maximal monotonicity is equivalent to $0 \in \text{ran}(A + \text{Id})$
2. **Minty's Theorem**: If A is maximally monotone, then $\text{ran}(\text{Id} + A) = H$
3. **Debrunner-Flor Theorem**: Provides a separation property for monotone sets and characterizes maximal monotone extensions
4. **Domain-Range Relations**: For maximally monotone A with $\text{int dom } A \neq \emptyset$: - $\text{dom } A = \overline{\text{dom } A}$ and $\text{int dom } A = \text{dom } A$ - Both $\text{dom } A$ and $\text{ran } A$ are convex

Key Theorems and Results II

5. **Local Boundedness**: (Rockafellar-Vesely, Theorem 21.18) - A is locally bounded at $x \iff x \notin$ bdry dom A
6. **Surjectivity Conditions**: - A surjective $\iff A^{-1}$ locally bounded everywhere - If $\lim_{\|x\|\to\infty} \inf \|Ax\| = +\infty$, then A surjective - If dom A bounded, then A surjective
7. **Kenderov's Theorem**: For maximally monotone A with int dom $A \neq \emptyset$: - There exists a dense G_δ subset C of dom A - On C: Ax is singleton and every selection is continuous
8. **Fréchet Differentiability**: Convex functions with nonempty interior domain are Fréchet differentiable on a dense G_δ subset

Connections to Other Chapters

- **Chapter 16**: Single-valued subdifferentials and continuous selections
- **Chapter 20**: Properties of convex functions (from which these results are applied)
- **Chapter 21**: Finer properties of monotone operators beyond basic characterizations
- **Chapter 22**: Stronger notions of monotonicity (para-, strict, uniform, strong monotonicity)
- **Applications**: Optimization algorithms, evolution equations, variational inequalities

Key Exercises from Chapter 21

- **Exercise 21.1**: Show that A is maximally monotone iff $\operatorname{gra} A + \operatorname{gra}(-\mathrm{Id}) = H \times H$
- **Exercise 21.2**: Matrix monotonicity (for symmetric matrices)
- **Exercise 21.3**: Distance to graph using Gâteaux differentiability
- **Exercise 21.5**: Equivalence of maximal monotonicity with: (i) Strong-to-weak continuity (ii) Hemicontinuity
- **Exercise 21.6**: Subdifferentials of Gâteaux differentiable functions
- **Exercise 21.12**: Finite-dimensional result on bounded subsets
- **Exercise 21.17**: Fréchet differentiability instances (at what sets of points)

Further Reading

- **Minty (1962)**: Original work on maximally monotone operators and their properties
- **Rockafellar (1970)**: Monotone operators and convex analysis
- **Bauschke & Combettes (2017)**: Comprehensive modern treatment with Hilbert space setting
- **Zarantonello (1971)**: Solving Functional Equations by Contraction Mappings
- **Brezis (1973)**: Opérateurs Maximaux Monotones et Semi-groupes de Contractions

Chapter 21: Finer Monotone Properties

Maximally monotone operators form a powerful framework for convex analysis and the study of operator equations.

The results presented here—Minty, Debrunner-Flor, Rockafellar-Vesely, and Kenderov—are fundamental tools for modern optimization and variational analysis.